

Check Sheet

Comprehensive Diagnostic Checklist

Introduction

Purpose of This Check Sheet

This checklist helps technicians carry out structured, consistent diagnostics. It covers the major systems, common fault areas, and key verification steps needed before repairs begin. Use it to confirm symptoms, rule out simple issues, and document findings clearly.

When to Use This Document

Use this sheet during:

- Initial vehicle inspection
 - Fault confirmation
 - Pre-repair verification
 - Post-repair validation
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Vehicle Information

Basic Details

- VIN
 - Registration number
 - Model and variant
 - Model year
 - Odometer reading
 - Date of inspection
 - Technician name
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Initial Checks

Customer Complaint Verification

- Confirm reported symptoms
- Clarify when the issue occurs (cold, hot, acceleration, braking, etc.)
- Note any unusual noises, smells, or performance changes
- Check for recent repairs or modifications

Warning Lamps and Messages

- Check instrument cluster for warning indicators
- Record active alerts or messages
- Confirm whether alerts appear intermittently or consistently

Visual Inspection

- Look for leaks (oil, coolant, fuel, hydraulic)
 - Check belts and hoses for damage
 - Inspect connectors for corrosion or loose pins
 - Verify battery connections
 - Confirm no aftermarket wiring is interfering with systems
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Electrical System Checks

Battery and Charging

- Measure battery voltage (engine off)
- Measure charging voltage (engine running)
- Check battery terminal condition
- Inspect ground points

Fuses and Relays

- Inspect relevant fuses for continuity
- Check relay operation
- Confirm correct fuse ratings

Wiring and Connectors

- Inspect wiring for abrasion or exposure
 - Confirm connector locking tabs are secure
 - Check for moisture ingress
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Diagnostic Tool Checks

DTC Scan

- Connect diagnostic tool
- Read and record all active and stored DTCs
- Capture freeze frame data when available
- Clear codes only after logging them

Live Data Monitoring

- Monitor key parameters related to the fault
- Check sensor values for plausibility
- Compare readings to specification ranges

Actuator Tests

- Run relevant actuator or output tests
 - Confirm components respond correctly
 - Document any failures or irregular responses
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Powertrain Checks

Engine Systems

- Check air intake for blockage
- Inspect air filter
- Verify throttle response
- Listen for abnormal engine noise
- Confirm coolant temperature stability

Fuel System

- Check fuel level and quality
- Inspect fuel lines
- Verify fuel pressure (if applicable)
- Look for injector imbalance

Exhaust System

- Inspect for leaks
 - Check condition of sensors (O2, NOx, etc.)
 - Note unusual smells
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Chassis and Mechanical Checks

Braking System

- Inspect brake pads and discs
- Check brake fluid level
- Look for leaks in brake lines
- Confirm ABS warning lamp status

Suspension and Steering

- Check for play or looseness
- Inspect bushings and joints
- Look for uneven tyre wear
- Confirm power steering operation

Drivetrain

- Inspect driveshafts
 - Listen for clicking or grinding noises
 - Check differential oil level (if applicable)
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HV and Electrical Subsystems

(For Electric and Hybrid Vehicles)

High-Voltage Safety

- Confirm HV system isolation
- Verify HV interlock integrity
- Inspect orange cables for damage
- Check insulation resistance (if procedures allow)

Battery System

- Record state of charge
 - Check cooling system operation
 - Review battery DTCs and temperature data
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Final Verification

Road Test

- Confirm symptoms under controlled conditions
- Note when issues occur (speed, load, temperature)
- Monitor live data during test
- Check for vibrations, noises, or hesitation

Post-Test Review

- Recheck for new DTCs
 - Compare pre-test and post-test values
 - Validate repairs (if performed)
 - Confirm all warning lamps are cleared
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Technician Summary

Findings

- List confirmed faults
- Note systems affected
- Identify root cause (if known)

Recommended Actions

- Outline repair steps
- Note required parts
- Estimate repair time

Completion

- Technician signature
- Supervisor approval (if required)